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Q1. String Character Frequency Counter

CODE

```
text=input("enter a string\n")
freq={}
for ch in text.lower():
    if ch!=" ":
        if ch in freq:
            freq[ch]+=1
        else:
            freq[ch]=1
for ch in sorted(freq):
    print(ch, "→",freq[ch])
```

INPUT

Hello World

OUTPUT

```
enter a string
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
text=input("enter a sentence:").lower()
v="aeiou"
vf={}
cf={}
vc=cc=0
for ch in text:
    if ch.isalpha():
        if ch in v:
            vc+=1
            vf[ch]=vf.get(ch,0)+1
        else:
            cc+=1
            cf[ch]=cf.get(ch,0)+1
print("vowels:", vc)
print("consonants:",cc)
if vf:
    x=max(vf,key=vf.get)
    print("most freq vowel:",x,"(",vf[x],")")
else:
    print("none")
if cf:
    y=min(cf,key=cf.get)
    print("least freq consonant:",y,"(",cf[y],")")
else:
    print("none")
print("vowel to consonant ratio:", round(vc/cc,2) if cc else "underdefined")
```

INPUT

Hello World

OUTPUT

```
enter a sentence:vowels: 3
consonants: 7
most freq vowel: o ( 2 )
least freq consonant: h ( 1 )
vowel to consonant ratio: 0.43
```

Q3. String Compression and Run-Length Encoding

CODE

```
def compress(s):
    r=""
    i=0
    while i< len(s):
        c=1
        while i<len(s)-1 and s[i]==s[i+1]:
            c+=1
            i+=1
        r+=s[i]+(str(c) if c>1 else "")
        i+=1
    return r if len(r)<len(s) else s
def decompress(s):
    r=""
    i=0
    while i<len(s):
        ch=s[i]
        i+=1
        n=""
        while i<len(s) and s[i].isdigit():
            n+=s[i]
            i+=1
        return r+ch*n
s = input()
c=compress(s)
if c==s:
    print("original:", s)
else:
    print("compresses:",c)
    print("decompressed:", decompress(c))
```

INPUT

aabbccdddee

OUTPUT

compresses: a2b3c2d4e2
decompressed:

Q4. Advanced Text Statistics Analyser

CODE

```
text=input("enter text:\n")
words=text.split()
sentences=text.count(".") + text.count("!") + text.count("?")
if sentences == 0:
    sentences=1
paragraphs=1
print("words:", len(words))
print("sentences:", sentences)
print("paragraphs", paragraphs)
print("average word length:", round(sum(len(w) for w in words)/len(words), 2))
print("average sentence length:", round(len(words)/sentences, 2))
stop = ["the", "is", "a", "an", "in", "of", "and", "to", "for"]
count={}
for w in words:
    w=w.lower().strip(".,!?")
    if w not in stop:
        count[w]=count.get(w, 0)+1
print("top 5 words:")
for w,c in sorted(count.items(), key=lambda x: x[1], reverse = True)[:5]:
    print(w, ":", c)
```

INPUT

python is powerful programming language. python is easy to learn. programming in python is fun!

OUTPUT

```
enter text:
words: 15
sentences: 3
paragraphs 1
average word length: 5.4
average sentence length: 5.0
top 5 words:
python : 3
programming : 2
powerful : 1
language : 1
easy : 1
```